Coleman Breen

Current Position

2026–present **Head of AI Policy**, *SecureBio*, Washington, DC

- Lead organization's US and European policy portfolio on AI and biosecurity
- Lead SecureBio's contribution to CBRN risk assessment and evaluation for the EU AI Office under the EU AI Act
- Design and oversee evaluations of frontier AI systems' biosecurity-relevant capabilities for policy-makers
- Brief US government agencies and congressional staff on frontier-model biological capabilities
- Author policy analysis for the AI governance community on AI-enabled biological risk

Education

2020–2027 **Ph.D. in Statistics**, *University of Wisconsin–Madison*
Expected Spring 2027. Advisor: Sündüz Keleş. Dissertation on sequence-based prediction of DNA methylation.

2023 **M.S. Statistics**, *University of Wisconsin–Madison*

2016–2019 **B.S. Statistics & Analytics; B.A. Mathematics**, *University of North Carolina at Chapel Hill*

Professional Experience

2025–2026 **Senior AI Policy Researcher; Research Scientist, AI Team**, *SecureBio*

- Led SecureBio's work supporting EU AI Act implementation, helping EU regulators measure AI capabilities for CBRN risk assessment
- Co-led ReproBAIT, a joint benchmark with the UK AI Security Institute testing whether coding agents can rebuild biological AI tools
- Led development of an expert-written benchmark of AI knowledge in agricultural biology
- Contributed to human-uplift and capability evaluations of frontier models, including pre-release assessments

2025 **Fellow, AI and Biosecurity Policy**, *Johns Hopkins Center for Health Security*, Washington, DC

- Analyzed policy issues at the intersection of artificial intelligence and biotechnology

Mar–Sep 2024 **AI and Plant Biology Resident**, *Google[x] / Heritable Agriculture*

- Developed biological AI models for an agriculture moonshot

2020–2025 **Research Assistant**, *University of Wisconsin–Madison*, Department of Statistics
Advised by Sündüz Keleş and Reid Alisch

- Developed statistical and deep-learning models for whole-genome methylation data (more than 25 million CpG sites) in Alzheimer's disease
- Built and maintained processing pipeline for a large genomic database
- Consulted on genotype associations with treatment response in age-related macular degeneration and on mitochondrial DNA methylation in platelet storage

2017–2020 **Undergraduate and Post-baccalaureate Researcher**, *University of North Carolina at Chapel Hill*, Maddox Lab, Department of Biology

- Modeled imaging data from cytokinesis with wavelets and oscillation analysis

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Publications

Peer-reviewed articles

2025 Madrid A, Papale LA, Bergmann PE, **Breen C**, Clark LR, Asthana S, Johnson SC, Keleş S, Hogan KJ, Alisch RS. Whole genome methylation sequencing in blood from persons with mild cognitive impairment and dementia due to Alzheimer's disease identifies cognitive status. *Alzheimer's & Dementia* 21(2):e14474. doi:10.1002/alz.14474

2024 **Breen C**, Papale LA, Clark LR, Bergmann PE, Madrid A, Asthana S, Johnson SC, Keleş S, Alisch RS, Hogan KJ. Whole genome methylation sequencing in blood identifies extensive differential DNA methylation in late-onset dementia due to Alzheimer's disease. *Alzheimer's & Dementia* 20(2):1050–1062. doi:10.1002/alz.13514

2024 Werner ME, Ray DD, **Breen C**, Staddon MF, Jug F, Banerjee S, Maddox AS. Mechanical and biochemical feedback combine to generate complex contractile oscillations in cytokinesis. *Current Biology* 34(14):3201–3214.e5. doi:10.1016/j.cub.2024.06.037

Preprints

2026 Muthupandiyan S, Dur TH, Walker M, Flesch T, Ghanem M, Suddell S, Harper-Donnelly G, Rose S, Götting J, **Breen C**, Pencharz J. ReproBAIT: Can coding agents build their own biological AI tools? *arXiv*:2610.19810. Joint work with the UK AI Security Institute.

2026 Zhang CBC, Knight CQ, Kruus N, Hausenloy J, Medeiros P, Li N, Kim A, Orlovskiy Y, **Breen C**, Cai B, Götting J, Liu AB, Nedungadi S, Rodriguez P, He YY, Shaaban M, Wang Z, Donoughe S, Michael J. LLM novice uplift on dual-use, *in silico* biology tasks. *arXiv*:2602.23329

In preparation

Breen C, Keleş S. Personalized sequence encoding does not reproducibly improve DNA methylation prediction across nine mouse strains.

Conference abstracts

2024 Bergmann PE, Madrid A, Papale LA, **Breen C**, et al. Whole genome methylation sequencing identifies sex-specific DNA methylation levels in late onset dementia due to Alzheimer's disease. *Alzheimer's & Dementia* 20(S1):e087422 (AAIC 2024).

2024 Boruch AE, Renteria I, Madrid A, **Breen CE**, Papale LA, et al. Whole genome methylation sequencing in blood reveals differential DNA methylation in pathways associated with mitochondrial function in persons with late-onset dementia due to Alzheimer's disease. *Alzheimer's & Dementia* 20(S1):e088336 (AAIC 2024).

2024 Madrid A, **Breen CE**, Papale LA, Clark LR, et al. Whole genome methylation sequencing in blood identifies extensive differential DNA methylation in mild cognitive impairment. *Alzheimer's & Dementia* 20(S1):e088557 (AAIC 2024).

2023 Werner ME, Ray D, **Breen C**, Jug F, Maddox A. Speed oscillations in cytokinetic ring ingression suggest interplay between mechanical and biochemical feedback. *Molecular Biology of the Cell* 34(2):102 (ASCB).

2020 Werner M, Ray D, **Breen C**, Sattler A, Jug F, Maddox AS. Harmonic oscillation frequencies of cellular contractility support a wave shape model. *Bulletin of the American Physical Society* 65.

Reports and policy writing

Oct 2026 Sudarshan B, Kara Z, van der Merwe M, Halstead J, Allen S, Bastians H, **Breen C**, Mahar M, Mayers C, McCaslin T, Nelson C. Managing biological risks from general-purpose AI models under the EU AI Act. *Centre for the Governance of AI (GovAI)*, Technical Report.

Oct 2026 Muthupandiyan S, Walker M, Götting J, **Breen C**. ReproBAIT: A benchmark for reproducing biological AI tools. *SecureBio*.

Jul 2026 **Breen C**, Omaar H. Preparing for the "Bio Mythos" moment. *SecureBio*.

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Invited Talks

Sep 2026 **Safety, Risk, and Adversarial Evaluation (invited remarks)**, *National Academies of Sciences, Engineering, and Medicine*, Washington, DC
Meeting of Experts on the Science of Evaluation of AI for Life Sciences and Biotechnology

Sep 2026 **State of AIxBio Guardrails (workshop lead)**, *Biosecurity Summit*, Lansdowne, VA

Jul 2026 **Opening remarks, AI–bio nexus session**, *CNAS U.S.–Northeast Asia Forum on AI Risks and Opportunities*, virtual
Track II dialogue

Mar 2026 **A Biosecurity Evaluations Ecosystem with Policymaking in Mind**, *FAR.AI Technical Innovations for AI Policy Conference*, Washington, DC

Mar 2026 **Beyond Expert-level Q&A: Biosecurity Evals for Increasingly Capable AI**, *Nexus Series AIxBio Workshop 2*, Johns Hopkins Bloomberg Center, Washington, DC
Hosted by Johns Hopkins University and RTX BBN

Nov 2025 **Statistical Best Practices for AI Capabilities Evaluations**, *Nexus Series AIxBio Workshop 1*, Johns Hopkins Bloomberg Center, Washington, DC
Hosted by Johns Hopkins University and RTX BBN

Earlier Presentations

Aug 2022 **American Society for Human Genetics**
Contributed poster

Jul 2022 **National Library of Medicine T32 Conference**
Contributed talk

Jul 2022 **Intelligent Systems for Molecular Biology**
Contributed poster

Jun 2022 **UW–Madison Neurological Surgery Research Day**
Lightning talk

Oct 2021 **UW–Madison Epigenetics Symposium**
Lightning talk

Oct 2021 **UW–Madison Neurological Surgery Research Day**
Lightning talk

Mentoring and Service

2027 **Mentor, SecureBio AI stream**, *ML Alignment & Theory Scholars (MATS) Program*
Winter 2027 cohort

2019 **Buckley Public Service Scholar**, *Carolina Center for Public Service*

2017–2019 **Senior Advocate**, *Community Empowerment Fund*

- Met one-on-one with individuals experiencing homelessness to help them obtain healthcare and housing

Skills

Computing Python (numpy, pandas, PyTorch, TensorFlow), R (data.table, tidyverse, ggplot2), Unix/Bash, GNU Parallel, bcftools, samtools, LLM evaluation frameworks

Methods Statistical modeling and inference, deep learning for genomic sequence, AI capability and safety evaluation, experimental design and power analysis

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